

**Overall model comparison for WEATHER**  
**Lookback = 168 steps, Forecast horizon = 24 steps**

| Model         | RMSE   | MAE    | sMAPE (%) | R <sup>2</sup> | Bias   | NRMSE |
|---------------|--------|--------|-----------|----------------|--------|-------|
| CNN           | 36.485 | 11.515 | 66.935    | 0.987          | -0.742 | 0.198 |
| LSTM          | 38.199 | 11.569 | 67.416    | 0.986          | 1.263  | 0.207 |
| dCeNN+ELM     | 43.531 | 14.939 | 71.824    | 0.982          | -1.464 | 0.236 |
| dCeNN+ELM+ASP | 42.440 | 13.281 | 53.453    | 0.983          | -0.980 | 0.230 |

- This table compares all models for one fixed experimental setting.
- For RMSE, MAE, sMAPE, and NRMSE, smaller values are better. For R<sup>2</sup>, larger values are better. For Bias, values closer to zero are better.
- Weather combines multiple physical variables. Aggregate RMSE or MAE should therefore be read mainly as a model-comparison tool, not as a single physical-unit error.